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Instruction Manual

APS 0109

Position Controller

S/N : 208

Original Instruction Manual

Imprint

Heidelberger Straße 12
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The entire documentation is not subject to any automatic updating. It reflects the present state of the art upon delivery. Contact the manufacturer for the current version.

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Table of Contents

1 Introduction

The type APS0109 Position Controller is for controlling the zero position of a vibration exciter (shaker). Its control characteristics can be adjusted, which makes it suitable for universal use. The Position Controller was developed especially for use in conjunction with air-bearinged shakers, but beyond that it can also be used for any other vibration exciter. It can be used on vibration exciters operating in horizontal direction, too.

The Position Controller consists of an optical sensor head and a control device containing a control unit and an actuating and monitoring unit. The actuating and monitoring unit is for implementing the following jobs:

- well-defined switching-on and from regime
- well-defined and soft shutting down when a malfunction (error) has occurred
- monitoring the maximum vibration displacement
- monitoring the air pressure of an air-bearinged shaker

The operating condition of the device and the conditions of protective functions are indicated by LEDs.

Type APS0109 can optionally be complemented

- with an interface for making connection to an SPC
- with an option for remote control and status query via RS 232

You can optionally configure the signal input as a floating or balanced input to avoid hum pick-up. An active hum suppressor can be inserted in the signal output chain.

The Position Controller can be combined with each of the following displacement measuring systems. Exchanging the type of displacement measuring system, however, requires the hardware of the APS0109 to be changed accordingly.

Options:

-G: use of light barriers with optical wedges as displacement measuring systems.

-L: use of a laser device as a displacement measuring system

-T: use of a triangulation sensor as a displacement measuring system

2 Specification

Operating voltage:	115 V (fuse: slow blow 630 mA / 250 V)
	230 V (fuse: slow blow 315 mA / 250 V)
	(100 V (fuse: slow blow 630 mA / 250 V) on request)
Maximum input voltage:	±15 V
Internal gain:	≤ 0 dB
Bandwidth:	0.1 Hz…25 kHz
Setting range of zero:	±10 % of maximum displacement
	(±20 % on request)
Weight (net weight)	3.9 kg (8.6 lbs)

3 General description

3.1 Principle of operation

The type APS0109 Position Controller is made up of the following component parts: optical sensor head and control device containing a control unit and an actuating and monitoring unit. The optical sensor head is mounted onto the shaker, i.e. detached from the control device. Its job is to monitor the instantaneous position of the shaker head and, by doing so, also supply the data needed for recognizing any possible exceeding of the maximum admissible vibration displacement (overtravel).

The controller is configured as a PI controller and so ensures that the zero position is kept constant even if the shaker load is varied. Control speed (stiffness) and zero position (zero) can be adjusted independently. .

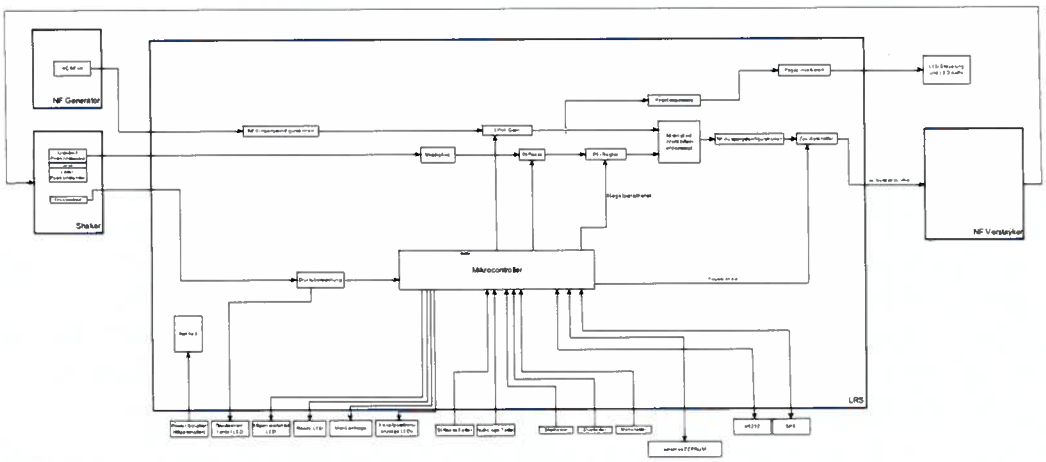
The control device generates a DC voltage offset (bias) that is superimposed on the signal for driving the shaker. For the Position Controller to work properly, you are required to use a power amplifier capable of processing DC voltage signals. The signal source (generator) is required to supply a signal that is free from any DC offset.

The operation of the Position Controller includes a well-defined switching-on and from regime. Thus the signal inputs will be connected through only after the zero position has been attained. If the Position Controller is switched off after reaching zero position, the signal inputs will be shut down first and the shaker will be moved safely into its position of rest.

The signal inputs will be shut down first when a malfunction (error) has occurred. Subsequently the shaker will be moved into its position of rest.

If after start-up (from the overtravel) the shaker head is unable to move within a specified period of time, the controller will change over to error condition. On systems with laser position sensor the time out observation will be active until the zero position is reached and stiffness is ramped to its final condition. There is an open-loop detector that will shut the device down if the control loop is interrupted (e. g. amplifier breakdown).

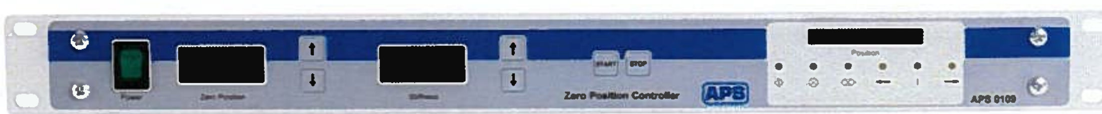
Pressure in the compressed air piping is monitored by means of a pressure sensor. You will not be able to start the controller without pressure. The pressure sensor has to be configured as a floating pressure switch. It is not part of the items supplied.



Block diagram of Position Controller LRS

3.2 Front panel

The front panel contains all controls and indicators of the Position Controller.



There are 3 LEDs for indicating the position of the shaker head in relation to the set zero position.

There are another 3 LEDs for indicating the state of operation of the device:

POWER LED (<|>): indicates that power is supplied to the APS0109.

Attention! If the POWER LED does not light up after the APS0109 has been started, the internal or external EEProm have read out illegal configuration data and all configuration data have been reset to the manufacturer's default settings.

It is absolutely necessary to set / check the parameters of the APS0109.

ERROR LED (⊗): - on: error; shaker is in position of rest

- flashing - settling time is running - shaker head moving into zero position

- off, full functionality - shaker is ready for use; signal inputs are connected through

PRESS. FAIL LED (): - on, no compressed air, or amplifier error, or cable to displacement sensor is not connected

- off, compressed air is available or no amplifier error, cable to displacement sensor is connected

3.3 Connections

Signal input and signal output are located in the rear of the device. The Position Controller has to be inserted into the signal path between generator and power amplifier.

The optical sensor head has to be connected (together with the pressure sensor) using the corresponding connector on the real panel.

The RS232 interface socket is located on the rear panel of the APS0109, too.

Sockets for optional connections to SPC and external memory module are also located on the rear panel of the APS0109.

3.4 Operation

Press key **Start** to start the controller. The shaker head will move into its zero position. Press key **Stop** to stop the controller. The signal inputs will be interrupted and the shaker will move into its position of rest.

There are two controls on the device with the following functions:

Stiffness - *Control speed (stiffness) of the controller* - The appropriate setting depends on the intended mode of operation of the shaker. Operating the shaker at low frequencies (< 10 Hz) will require a smaller control speed than at higher frequencies.

Zero - *Zero position of the shaker head* - If the knob is in its vertical position, the shaker is in its mechanical zero position. The zero position of the head can be shifted in a range of approx. ± 10 % of maximum vibration displacement around the mechanical zero position.

3.5 Version Firmware

The version of the firmware will be displayed for 2 seconds after power source is switched the bootloader loaded and the display test was running.

If the Zero display shows 0 and the Stiffness display shows 5 then the firmware version is 05.

4 Getting started

The detailed steps for the startup configuration please look at the startup manual.

4.1 Polarity check

Make all electrical connections, switch the compressed air on and switch the power amplifier on. After that, press key **Start** to switch the controller on. Observe the head. If it does not move into its zero position but instead is pulled into the magnet, invert the polarity of the shaker.

This can be done either by inverting the shaker polarity at the output of the power amplifier or by changing the corresponding setting of the device in Menu 12 (polarity inversion of the output signal).

4.2 Matching with power amplifier

The gain factor of the employed power amplifier has an effect on the control characteristic of the APS0109.

To ensure flawless operation of the controller, move to Menu 17 to set the parameters of the APS0109. Depending on the internal gain of the power amplifier, the following settings should be selected:

Menu entry	Gain of APS0109	Gain of power amplifier	Power output of amplifier (1 V input)	Power output of amplifier (10 V input)	k constant factor for determining the transfer coefficient
2	4	2.5	*	150 VA	12 V
2	3	3	*	250 VA	9 V
3	2	5	6 VA	600 VA	6 V
6	1	10	25 VA	2500 VA	3 V
10	0.66	15	60 VA		2 V
14	0.5	20	100 VA		1.5 V
21	0.33	25	200 VA		1.1 V
25	0.28	35	300 VA		0.88 V
28	0.25	40	400 VA		0.75 V
42	0.166	60	1000 VA		0.5 V
55	0.125	80	1500 VA		0.325 V

Attention!

To avoid potential differences of the shaker in the protective conductor you must run the power amplifier in the same power line as the positioning controller.

In case this is not possible it is prescribed to use the option "buffer amplifier".

The stated values of power output can be used as a rough basis for assessment in case the internal gain of the power amplifier is not known.

4.3 Configuration of input and output sockets

The configuration of the input socket can be set by making the corresponding entry in Menu 11 as shown below.

The configuration of the output socket can be set by making the corresponding entries in Menu 12 and in Menu 13. Disturbing influences due to hum pick-up can be reduced effectively by appropriately configuring the sockets.

The following configurations can be selected:

Menu 11: input configuration:

- 0: floating
- 1: relative to earth

Menu 13: output configuration:

- 0: output signal relative to earth
- 1: active hum suppression

4.4 Operation without using compressed air

If the employed shaker is designed for use without compressed air, the compressed air monitor can be switched off by changing the corresponding entry in Menu 9. In this case the **Pressure** LED will be permanently off.

4.5 Adjusting the mechanical zero position

The optical sensor head has to be attached to the shaker such that the head is in its mechanical zero position when the **Zero** control is set to zero. The mechanical zero position of the bar graph can be fine adjusted in Menu 27 and Menu 28.

The sensor head position has to be adjusted coarsely on the sensor head. It can be checked in 53 and 54 in case no voltage meter is available.

For fine adjustment navigate to 36 and 37.

The fine adjustment can also be performed on the APS0109:

Requirements: the bar graph is adjusted, the settings for the amplifier / control parameters are founded, the APS0109 is ready to start.

Execution:

Start the APS0109 and wait for the alignment of the zero position.

After this, adjust the correct zero position using the zero keys.

Press the start key till the APS0109 switch off the control of the zero position (about 5 seconds).

The value founded can be changed in 36.

4.6 Using the APS 0109 on modal shakers

If the APS0109 is used on modal shakers for positioning and generation of a bias force, the controller has to be configured as a P (proportional) controller.

To this end, enter value 0 in Menu 18.

4.7 Operation as a P controller

If the APS0109 is used as a P (proportional) controller (Menu 18, value 0), the controller can be looked at as a spring-damper combination, the transfer coefficient *h* (force constant) of which can be adjusted by means of the stiffness control. If the stiffness control is set to full scale, the transfer coefficient *h,* depending on the selected amplifier matching, is:

$$h = \frac{k}{s_{\max}}$$

where

k: constant factor according to Table 4.1

s_{max}: maximum displacement of vibration exciter

Attenuation can be adjusted in Menu 20 to adapt the system to the current application. To avoid any reaction on the shaker, you should navigate to Menu 20 2, frequencies, select <10Hz and increase attenuation. In doing so, please be aware that the correct setting also depends on the employed shaker.

5 Remote control functions

The remote control functions that are available as an optional extra allow the Position Controller to be controlled and monitored by some external device.

5.1 Linkage to SPS/Amplifier

The SPC interface has 24V compatible inputs and outputs.

Output 1 is for indicating whether the zero position has been reached.

Attention!

The output 1 is directly connected to the green LED for the indication of the zero position.

The output 1 will not signaling, if the controller has finished the positioning of the shaker head or not. It only will show the zero position.

The positioning has finished after the zero position is reached, the stiffness is ramped to the final value and the input signal is patched through. After that, the flashing of the **ERROR LED (⊗) will stop.**

Output 2 is for signaling error condition Overtravel.

Output 3 is for signaling error condition No compressed air.

Output 4 is for signaling the stop condition.

Input 1 is for controlling the controller. H level will start the controller, L level will stop it. Each time an error has occurred, this input has to be set to L level for at least 20 msec, after that the system can be restarted.

Input 2 is for shutting down the control action. This is used to control the amplifier reset signal.

Input 3 is for shutting down the control action. This is used to control the amplifier interlock signal.

Input 4 are kept in reserve for further applications.

Pin assignment of socket for link to SPC:

SPC	Sub-D_25Pol_Buchse
1	SPS ina_1
2	SPS inb_1
3	SPS ina_2
4	SPS inb_2
5	SPS ina_3
6	SPS inb_3
7	SPS ina_4
8	SPS inb_4
9	Out Logo
10	AGND
11	-
12	-
13	-
14	SPS outa_1
15	SPS outb_1
16	SPS outa_2
17	SPS outb_2
18	SPS outa_3
19	SPS outb_3
20	SPS outa_4
21	SPS outb_4
22	AGND
23	AGND
24	-
25	-

The Out Logo connector returns a position signal of the head.

When using a displacement measuring system with light barriers, this signal will be temperature dependent.

When using a triangulation sensor or a laser sensor, this signal will not be temperature dependent.

5.2 Remote control via RS 232

Control via the serial interface (RS 232) is by far more efficient. The interface of the control device has to be configured for a speed of 19200 Baud (1 start bit, 1 stop bit, no parity).

The following commands have been implemented, every one of which needs to be concluded with $0:

RS232 Commands

6 RS232 commands

All commands for communication with the APS0109 consist of at least 3 letters.
The reply signal to a command is a repetition of the command, followed by OK if no error has been detected.
If some error has been detected, e. g. exceeding of range, the reply signal is ERR.

The error condition can be queried using command GES.

6.1 ZER

ZERoposition

For setting the current value of the register for driving the zero position (indication + DAC value)
Syntax: **ZER Value**

Command: **ZER**
Value: **9** (decimal)

Maximum range of values: -99 to 99

Example: **ZER 9**

Replay from APS0109: **ZER OK**

6.2 ZER

ZERoposition reading (?)

For reading the current value from the register for driving the zero position (indication + DAC value)
Syntax: **ZER?**

Command: **ZER?**

Example: **ZER?**

Reply from APS0109: **ZER? 10 OK**

10 is the current value of the register in decimal representation

6.3 STF

STiFfness

For setting the current value of the register for driving stiffness (indication + DAC)
Syntax: **STF Value**

Command: **STF**
Value: **15** (decimal)

Maximum range of values: 0 to 31

Example: **STF 15**

Reply from APS0109: **STF OK**

6.4 STF?

STiFfness reading (?)

For reading the current value from the register for driving stiffness
Syntax: **STF?**

Command: **STF?**

Example: **STF?**

Replay from APS0109: **STF? 16 OK**

16 is the current value of the register in decimal representation

6.5 STA

STArt

For triggering the event Start in the controller.

Syntax: **STA Value**

Command: **STA**

Reply from APS0109: **STA OK**

6.6 STA?

STArt reading (?)

For returning a message whether control has been started (1) or the controller is in its state of initialization (0)
Syntax: **STA?**

Command: **STA?**

Example: **STA?**

Reply from APS0109: **STA? 1 OK**

16 is the current value of the register in decimal representation

6.7 STP

SToP

For triggering the event Stop in the controller.
Syntax: **STP**

Reply from APS0109: **STP OK**

6.8 STP

SToP reading (?)

For returning a message whether control has been stopped (1).
Syntax: **STP?**

Command: **STP?**

Example: **STP?**

Reply from APS0109: **STP? 1 OK**

16 is the current value of the register in decimal representation

6.9 FWV?

FirmWareVersion reading (?)

For reading out the firmware version of the APS0109.

Syntax: **FWV?**

Reply from APS0109: **FWV? 1 OK**

6.10 RST

ReSeT

For triggering a total reset of the APS0109. Attention, all settings not saved so far will get lost!

Syntax: **RST**

Reply from APS0109: **RST OK**

6.11 GES

Get Error String

For returning the last error condition in communication in the form of a string.

The following messages are implemented:

1. no_error -> no error has occurred
2. value_out_of_range -> number read out is out of validity range
3. unknown_command -> command has not been recognized
4. CRC_ERROR -> check sum in configuration block is wrong
5. Overtravel_TOP -> upper displacement range has been exceeded
6. Overtravel_Bottom -> lower displacement range has been exceeded

After the error string has been queried, it will be reset to no error.

6.12 SER?

SERial reading (?)

For reading out the serial number of the APS0109..

Syntax: **SER?**

Command: **SER?**

Maximum range of values: 0 to 65535

Example: **SER?**

Reply from APS0109: **SER? 1002 OK**

6.13 OTT

OverTravel Tolerance

For setting the value of tolerance of overtravel detection. This value is stored on the internal EEP and will be reloaded from the internal EEProm each time the APS0109 is started.

This value is subtracted from the value for upper overtravel recognition and added to the value for lower overtravel recognition.
Attention! Value is not mapped but directly related to the 10 Bit ADC.

Syntax: **OTT Value**

Command: **OTT**
Value: **15** (decimal)

Maximum range of values: 0 to 1023

Example: **OTT 15** //The tolerance band for detecting overtravel is 15 ADC values wide.
Reply from APS0109: **OTT OK**

6.14 OTT?

OverTravel Tolerance reading (?)

For reading out the value of tolerance of overtravel detection.

Syntax: **OTT?**

Maximum range of values: 0 to 1023

Example: **OTT?**

Reply from APS0109: **OTT? 15 OK**

15 is the current value of tolerance of overtravel detection.

6.15 PMA

Position MAximum

For setting the maximum value for driving the LEDs of the position indicator.

This value is saved on the internal EEProm of the APS0109.

If the position of the shaker head is greater than this value, the LED indicating the upper position of the head will be driven.

Attention, this value corresponds to the ADC value!

Syntax: **PMA Value**

Command: **PMA**
Value: **520** (decimal) default 459

Maximum range of values: 0 to 1023

Example: **PMA 520**
Reply from APS0109: **PMA OK**

6.16 PMA?

Position MAximum reading (?)

For reading out the maximum value for driving the LEDs of the position indicator.

If the position of the shaker head is greater than this value, the LED indicating the upper position of the head will be driven.

Attention, this value corresponds to the ADC value!

Syntax: **PMA?**

Command: **PMA?**

Example: **PMA?**
Reply from APS0109: **PMA? 520 OK**

6.17 PMI

Position MInimum

For setting the minimum value for driving the LEDs of the position indicator.
This value is saved on the internal EEProm of the APS0109.

If the position of the shaker head is smaller than this value, the LED indicating the lower position of the head will be driven.

Attention, this value corresponds to the ADC value!

Syntax: **PMI Value**

Command: **PMI**
Value: **480** (decimal) default 419

Maximum range of values: 0 to 1023

Example: **PMI 480**
Reply from APS0109: **PMI OK**

6.18 PMI?

Position MInimum reading (?)

For reading out the minimum value for driving the LEDs of the position indicator.

If the position of the shaker head is less than this value, the LED indicating the lower position of the head will be driven.

Attention, this value corresponds to the ADC value!

Syntax: **PMI?**

Command: **PMI?**

Maximum range of values: 0 to 1023

Example: **PMI?**
Reply from APS0109: **PMI? 480 OK**

6.19 SRV

SeRVice Mode

For switching the APS0109 into the Service Mode, so that the limits of overtravel of the second light barrier can be modified on the setting menu.

Syntax: **SRV Value**

Command: **SRV**
Value: **1**

Maximum range of values: 0 or 1 , 0 Service Modus is shut down, 1 Service Modus is enabled

Example: **SRV 1**

Reply from APS0109: **SRV OK**

6.20 SRV?

SeRVice Mode reading (?)

Reads out the status of Service Mode from the APS0109.

Syntax: **SRV?**

Command: **SRV?**

Example: **SRV?**

Reply from APS0109: **SRV? 1 OK**

Service Mode in APS0109 is enabled.

6.21 SSS

Set Stiffness Start Value

For setting the start value of stiffness of the APS 0109, after controlled operation until zero position has been reached.

Syntax: **SSS Wert**

Command: **SSS**
Value: 31

Maximum range of values: 0 to 31, default 15

Example: **SSS** 31

Reply from APS 0109: **SSS** OK

6.22 SSS?

Set Stiffness Start Value reading (?)

For reading from the APS 0109 the the value of stiffness that is set for reaching the zero position after controlled operation has been finished .

Syntax: **SSS?**

Command: **SSS?**

Example: **SSS?**

Reply from APS 0109: **SSS? 31 OK**

6.23 WTR

Wait Time Ramping

For setting the time for starting ramping the stiffness of the APS 0109.

Syntax: WTR **Value**

Command: **WTR**
Value: **100**

The time for starting ramping is calculated as follows: time in msec = 10 ms * value + 3500

Maximum range of values: 0 to 1000 , default 100

Example: **WTR 100** //1 second for ramping

Reply from APS 0109: **WTR OK**

6.24 WTR?

Wait Time Ramping reading (?)

For reading out the time for starting ramping the stiffness of the APS 0109.

Syntax: **WTR?**

Command: **WTR?**

Example: **WTR?**

Reply from APS 0109: **WTR? 100 OK**

7 Configuration menu

The configuration menu is for viewing and modifying the settings of the APS0109.

The configuration can be saved on the internal EEProm or the external EEProm, if enabled.

The APS0109 has to be in its Stop mode, otherwise you cannot enable the configuration menu.
First press key "Stop" (keep it depressed) and after that press key "Start".
Press both keys jointly for approx. 2 seconds.

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After that, the zero indicator will read out value 1.

Press key "Stop" for approx. 4 seconds if you want to leave the menu quickly from any point.
Attention! All settings made on the menu so far will get lost in this case!

Use the key for setting the zero position (Zero) for navigating through the main menu.
The indicator labeled Stiffness will read out the current value of the menu item.

To save the settings use menu 39 or 40.

Overview:

	Menu	Description	min val	max val	Comment	factory setting
1	1	For selecting the couple of values of Overtravel limits	1	3	1: selects limits on Menus 2 and 3 2: selects limits on Menus 4 and 5 3: selects limits on Menus 6 and 7	1
58	2	Overtravel limit couple 1 bottom	-99	99		-20
-18	3	Overtravel limit couple 1 top	-99	99		80
-20	4	Overtravel limit couple 2 bottom	-99	99		-20
80	5	Overtravel limit couple 2 top	-99	99		80
-20	6	Overtravel limit couple 3 bottom	-99	99		-20
80	7	Overtravel limit couple 3 top	-99	99		80
40	8	TimeOut of control action	1	99	sensor the time out observation will be active until the zero position is reached	10
0	9	Pressure or displacement sensor cable state monitoring (>= FW 0.20)	0	1	0: monitoring disabled 1: monitoring enabled	0
0	10	Horizontal shaker	0	1	0: no horizontal shaker, 1: horizontal shaker	0

0	11	For configuring the LF input	0	2	0: floating input 1: input signal relative to earth	0
1	12	For inverting the output signal	0	1	0: output not inverted 1: output inverted	0
1	13	Output signal relative to earth or active hum suppression	0	1	0: relative to earth 1: active hum suppression	0
1	14	Shaker without springs	0	1	0: shaker with springs 1: shaker without springs	1
5	15	Value for re-setting the step duration of updating the DC voltage in controlled mode of operation; value * 10 msec	0	99	has an effect on the speed of control	0
4	16	Value for setting the step width in controlled mode of operation	1	99	has an effect on the speed of control	1
25	17	Amplifier adjustment	0	63	Attention! Always start from high values (from 30)	60
1	18	For setting the I component of control (time constant)	0	6	0: no I component 1..6 selects a time constant	2
1	19	For limiting the I component	0	2	0: no limitation 1: limited to 1 Volt 2: limited to 2 Volt	0
1	20	P component	0	7	0..7 selects a gain	1
31	21	Attenuation of LF signal	0	31	0: strong attenuation 31: no attenuation	0
1	22	Which value of stiffness to use	0	1	0: the stiffness value set on the front panel is used as the default setting 1: the stored stiffness value is used as the default setting	1
4	23	Stored stiffness value	0	31		31
0	24	Which value of zero position to use	0	1	0: the zero position value set on the front panel is used as the default setting 1: the stored zero position value is used as the default setting	1
0	25	Stored zero position value	-99	99		0
i	26	For inverting the bar graph	0	1	0: do not invert 1: invert	0
53	27	For limiting the bar graph	0	63		63
13	28	Fine adjustment of zero position of bar graph	0	31		22
1	29	For selecting the signal source for the bar graph	0	1	0: input Overtravel A 1: input Overtravel B	0
0	30	For using SPC Remote	0	1	0: do not use SPC Remote control 1: use SPC Remote control	1

1	31	For using RS232	0	1	0: do not use RS232 interface 1: use RS232 interface	1
1	32	For using the front panel	0	1	0: do not use front panel 1: use front panel	1
1	33	For amplifier control	0	1	0: the amplifier signal reset and interlock are not checked 1: the amplifier signal reset and interlock are checked	1
-2	34	minimum value of zero position recognition	-99	99	for driving the LED	-1
3	35	maximum value of zero position recognition	-99	99	for driving the LED	2
-18	36	Zero value for fine adjustment of sensor head	-99	99	Attention! Until FW < 06 -99 = displacement of 0, value corresponding to the mapped ADC value of 0 to 1023, added to zero value Until FW >=06 Value is added direct to zero value e.g. zero value = 0 -> zero = zero + this value	0
0	37	Stiffness value for fine adjustment of sensor head	0	0	not used	0
0	38	For resetting to manufacturer's default settings	-	-	Pressing the "Stop" key will reset parameters	
0	39	For saving the settings on the APS0109	-	-	Pressing the "STOP" key will save the settings on the internal EEProm of the APS0109	
	40	For saving the settings on the internal and external EEProms	-	-	Pressing the "STOP" key will save the settings on the internal and external EEProms	
	41	Leave menu without saving	-	-		
	42	For indicating voltage Overtravel input A	-99	99		
	43	For indicating voltage Overtravel input B	-99	99		
	44	For indicating position input	-99	99		
	45	Query of Status input 1 for SPC (PC0)	0	1	indication only	
	46	Query of Status input 2 for SPS (PC1)	0	1	indication only	
	47	Query of Status input 3 for SPS (PC2)	0	1	indication only	
	48	Query of Status input 4 for SPS (PC3)	0	1	indication only	

49	Set output 1 for SPC (PC4)	0	1		
50	Set output 2 for SPC (PC5)	0	1		
51	Set output 3 for SPC (PC6)	0	1		
52	Set output 4 for SPC (PC7)	0	1		
53	Voltage adjustment of displacement measuring system input A	-1	+1	-1 voltage needs to be increased +1 voltage needs to be reduced	
54	Voltage adjustment of displacement measuring system input B	-1	+1	-1 voltage needs to be increased +1 voltage needs to be reduced	

7.1 Menu 1

For selecting the couple of values of overtravel limits

Advice:

If the position controller is working with an gray wedge and the overtravel limits are controlled by the wedge it is only possible to choose the voltages to identify the overtravel.
A adjustment of the displacement on these systems is not possible!

Possible values:

1: will select limits in Menu 2 and Menu 3
2: will select limits in Menu 4 and Menu 5
3: will select limits in Menu 6 and Menu 7

7.2 Menu 2

Minimum overtravel limit of overtravel value couple 1

Possible values: -99 to 99

7.3 Menu 3

Maximum overtravel limit of overtravel value couple 1

Possible values: -99 to 99

If this value and the value of signal 2 are exceeded, zero position control will be shut down and the output will be switched off.

7.4 Menu 4

Minimum overtravel limit of overtravel value couple 2

Possible values: -99 to 99

7.5 Menu 5

Maximum overtravel limit of overtravel value couple 2

Possible values: -99 to 99

If this value and the value of signal 2 are exceeded, zero position control will be shut down and the output will be switched off.

7.6 Menu 6

Minimum overtravel limit of overtravel value couple 3

Possible values: -99 to 99

7.7 Menu 7

Maximum overtravel limit of overtravel value couple 3

Possible values: -99 to 99

If this value and the value of signal 2 are exceeded, zero position control will be shut down and the output will be switched off.

7.8 Menu 8

TimeOut in seconds for controlled operation.
Check if the shaker head has moved within the specified period of time. If it has not moved, the settling procedure will be shut down. .

Possible values: 1 to 99

7.9 Menu 9

For switching pressure monitoring or displacement sensor cable state monitoring on / off

For displacement sensor cable state monitoring there must be a connection in the connection box (displacement sensor connection to APS 0109) between GND and the pressure signal (>= firmware version 0.20).

Possible values:
- 0: monitoring not enabled
- 1: monitoring is enabled

7.10 Menu 10

For setting if the shaker is operated in horizontal direction; has an effect on the switching-off characteristic.
If the shaker is operated in horizontal direction, the DC control signal will not be ramped down when switching off.

Possible values:
0: no horizontal shaker
1: horizontal shaker

7.11 Menu 11

For configuring the LF input, floating (0) or relative to earth (1)

Possible values:
- 0: floating
- 1: related to earth

7.12 Menu 12

For inverting the output signal

Possible values:
- 0: not inverted
- 1: inverted

7.13 Menu 13

Output signal relative to earth or active hum suppression

Possible values:
- 0: Output signal related to earth
- 1: active hum suppression

7.14 Menu 14

Shaker with or without springs

Possible values:
- 0: shaker with springs
- 1: shaker without springs

7.15 Menu 15

Value for setting the time of one step of updating the DC voltage in controlled mode of operation
Time = value * 10 ms.

Possible values: 0 to 99

0 is the fastest setting, 99 is the slowest setting

7.16 Menu 16

Value for setting the DC increment in controlled mode of operation.

Possible values: 1 to 99

Corresponding to the steps taken by the DAC, will take much time for value=1, 99 DAC will be set at once for value=99.

7.17 Menu 17

Amplifier matching

Possible values: 0 to 63

Attention! This setting is critical since the shaker may be damaged if a wrong setting has been selected.
It is absolutely necessary to make or check this setting after matching with power amplifier, and in doing so, always start with high values.

7.18 Menu 18

For setting the I component of control (time constant)

Possible values:

0: no I component
1…6 selects a time constant

1: TN = 1.06 s
2: TN = 0.80 s
3: TN = 0.46 s
4: TN = 0.32 s
5: TN = 0.24 s
6: TN = 0.19 s

7.19 Menu 19

For limiting the I component

Possible values:

0: no limitation
1: limitation to 1 Volt
2: limitation to 2 Volt

7.20 Menu 20

P component

Possible values:

0: KP = 1.2
1: KP = 0.8
2: KP = 0.6
3: KP = 0.5

4: KP = 0.4
5: KP = 0.35
6: KP = 0.3
7: KP = 0.25

7.21 Menu 21

Attenuation of LF signal

Possible values: 0 to 31

0 strong attenuation, 31 no attenuation, AC input signal is supplied to output without any attenuation.

7.22 Menu 22

For enabling the set stiffness value

Possible values:
- 0: the value set on the front panel will be used
- 1: the stored default value will be used

7.23 Menu 23

Stored value of stiffness (force constant)

Possible values: 0 to 31

7.24 Menu 24

For enabling the set zero position value

Possible values:
- 0: the value set on the front panel will be used
- 1: the stored default value will be used

7.25 Menu 25

Stored zero position value

Possible values: -99 to 99

7.26 Menu 26

For inverting the bar graph

Possible values:
- 0: do not invert bar graph
- 1: invert bar graph

7.27 Menu 27

For limiting the bar graph

Possible values: 0 to 63

For specifying the indicating range of the bar graph.

7.28 Menu 28

Fine adjustment of zero position of bar graph

Possible values: 0 to 31

Can be used for fine-tuning the zero position of the bar graph if it is not located directly at the center.

7.29 Menu 29

For selecting the bar graph signal

Possible values: 0 to 1

For selecting the signal source (input Overtravel A or Overtravel B) for the bar graph, if the bar graph does not continuously indicate the position. .
Attention! This setting also concerns the output of the positioning signal.

7.30 Menu 30

For using SPC Remote. This setting can be enabled only if the SPC is available to the APS0109.

Possible values:
- 0: Do not use SPC Remote
- 1: Use SPC Remote

Attention! If this option is selected and there is no SPC connected to the APS0109, you will not be able to start the APS0109 manually or via RS232, since the APS0109 will immediately detect Stop from the SPC and so shut down the control action.

7.31 Menu 31

For using the RS232, this setting has effect only if the APS0109 is equipped with optional extra Remote control via RS232 .

Possible values:
- 0: Do not use RS232
- 1: Use RS232

7.32 Menu 32

For enabling the front panel; specifies whether the keys on the front panel may be used for controlling the APS0109.
However, the configuration menu will accessible all the time.

Possible values:
- 0: Do not use front panel
- 1: Use front panel

7.33 Menu 33

For enabling the check for the power amplifier signals.

- 0: the amplifier signal reset and interlock are not checked
- 1: the amplifier signal reset and interlock are checked

Default 1

7.34 Menu 34

Minimum value for identifying the zero position of the shaker head (will be used for driving the LED of the shaker head)

Possible values: -99 to 99

See also Menu 44

7.35 Menu 35

Maximum value for identifying the zero position of the shaker head (will be used for driving the LED of the shaker head)

Possible values: -99 to 99

See also Menu 44

7.36 Menu 36

Zero value for fine adjustment of sensor head

Possible values: -99 to 99

7.37 Menu 37

Stiffness value for fine adjustment of sensor head

Not used

7.38 Menu 38

For using the "Stop" key for resetting to manufacturer's default settings

Possible values: -

7.39 Menu 39

For using the "Stop" key for saving the settings on the APS0109; values will be stored only on the internal memory of the APS0109.

Possible values: -

7.40 Menu 40

For using the "Stop" key for saving the settings on the internal and external memories
Attention! External memory must be available and the APS0109 must support this option.

Possible values: -

7.41 Menu 41

For using the "Stop" key for leaving the menu without saving

Possible values: -

7.42 Menu 42

Indication of voltage Overtravel input A

-99 to +99

7.43 Menu 43

Indication of voltage Overtravel input B

-99 to +99

7.44 Menu 44

Indication of Position input.

-99 to +99

7.45 Menu 45

Query of status input 1 for SPC

7.46 Menu 46

Query of status input 2 for SPC

7.47 Menu 47

Query of status input 3 for SPC

7.48 Menu 48

Query of status input 4 for SPC

7.49 Menu 49

For setting output 1 for SPC

7.50 Menu 50

For setting output 2 for SPC

7.51 Menu 51

For setting output 3 for SPC

7.52 Menu 52

For setting output 4 for SPC

7.53 Menu 53

Voltage adjustment displacement measuring system input B

Possible indications:
- -1: voltage < 8 Volt, voltage needs to be increased to achieve correct adjustment
- 1: voltage > 8 Volt, voltage needs to be reduced to achieve correct adjustment

7.54 Menu 54

Voltage adjustment displacement measuring system input B

Possible indications:
- -1: voltage < 8 Volt, voltage needs to be increased to achieve correct adjustment
- 1: voltage > 8 Volt, voltage needs to be reduced to achieve correct adjustment

8 Configuration with PC

The APS0109 can be configured using the PC software "APS0109 Config" that can be supplied by the manufacturer on request.
This software allows the configuration of the APS0109 to be read out from the APS0109 and to modify and again store a changed configuration.

For how to proceed, please refer to the Help menu of the software.

9 How to update the firmware

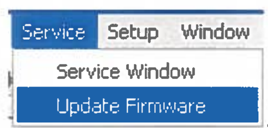
9.1 How to flash with the existing bootloader

Type APS0109 is equipped with a built-in boot loader that is enabled for 3 seconds immediately after the device has been switched on.

This boot loader will help you to load another firmware onto the APS0109 without the need to submit the device to the manufacturer.

Attention! At firmware loading, never switch the APS0109 off. Otherwise a broken firmware will be in the APS0109 and the APS0109 is not working!

9.1.1 Update with APS 0109 Config

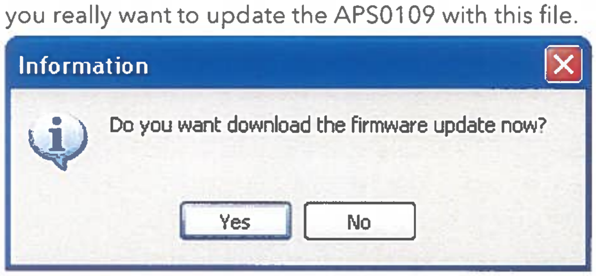
To update the firmware choose in menu 

.

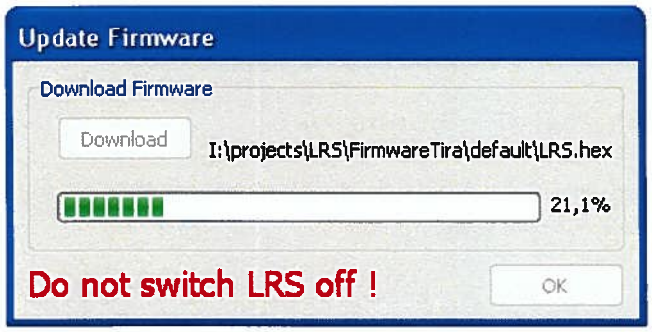
A window for the update will be displayed.

After pressing the button "Download" a dialog is displayed to select the firmware file. The file must be in the intel hex (*.hex) format.

After selecting the file and closing the dialog, if it is an valid firmware file you will be asked if you really want to update the APS0109 with this file.



If the dialog is closed by Yes, the update started



After successfully update the APS0109 will be restarted and the window for the update can be closed.

If there was an error at the update, this message will be displayed

The data and the power connection should be checked. May be it is necessary to restart the APS0109.
A new update can be started directly.

9.1.2 Update with BLIPS

Switch APS0109 off, start software BLIPS 2.6, select appropriate comport.
Use switch "Change Folder For AVR Data in XML" to select the BLIPS directory.
Make a RS232 connection to APS0109, in BLIPS set Baud rate to 19200
In BLIPS load flash file APS0109.hex, switch Auto-Verify after writing off
Use key "Flash Erase, Write" to start the system and immediately after that, switch the APS0109 on. If everything is OK, you should see the following image on your screen:

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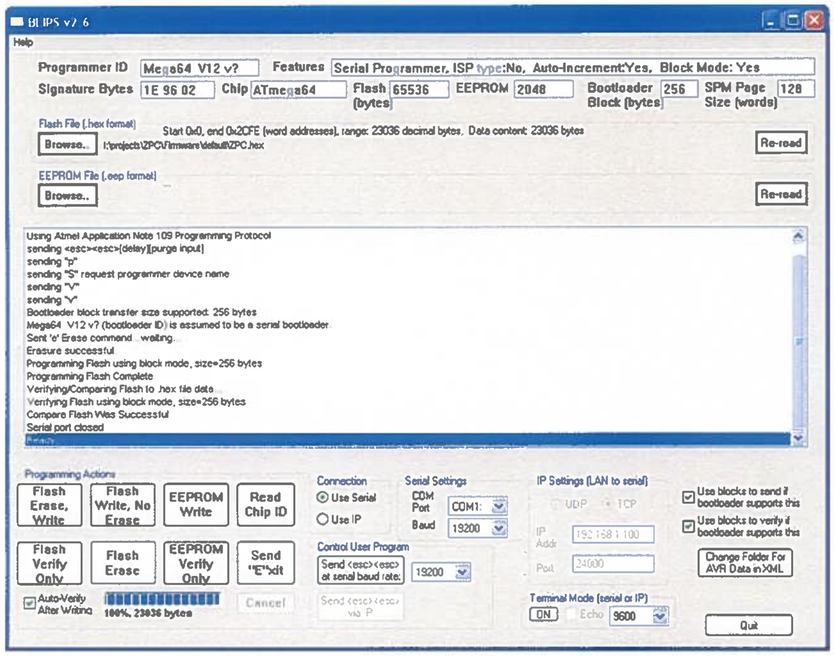

The new firmware will be enabled after restarting the APS0109.

10 Maintenance and repair

The type APS0109 Position Controller does not contain any wearing parts. Please consult the manufacturer and, if need be, submit the device to the manufacturer if you encounter a malfunction.

11 Warranty

The warranty of the product is contractually regulated.

SPEKTRA shall assume no liability for any damages and hazards that arise because the instruction manual or the safety warnings are disregarded.

In particular, the following actions will void the warranty:

- Execution of activities by incorrect group of personnel
- Improper installation of the product
- Insufficient cleaning
- Opening of the product
- Repair of the product
- Replacement of parts or cables
- Modification of the product
- Disassembly of the product

12 Contact

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